# Engineering Design

# Regolith Mars Rover Repeller

**- 2026 -**

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## Identification & Definition of Problem

Mars rovers operate in one of the harshest environments known to engineered systems. Rover wheel motors are crucial for mobility. These motors navigate uneven terrain, facing oxidized iron rocks, impact craters, and even volcanoes, whilst having to position scientific instruments for data collection. One of the most significant challenges facing the current Mars rover wheel systems are the fine Martian sand and regolith getting caught in wheel axles and motor assemblies. This issue creates a serious threat to rover longevity and mission success. Martian regolith is composed of extremely fine and abrasive particles that differ significantly from Earthen soil. These particles are small enough to penetrate narrow gaps within mechanical systems. Over time, dust accumulation inside motors of wheels increases friction and interferes with lubrication. As a result, this increases the rate of wear on bearings and other moving components, interfering with the productivity of the rover as a whole. Ideally, these rovers must function for years without maintenance. However, the gradual deterioration can lead to reduced efficiency and loss of mobile functionality altogether. The environmental conditions on Mars further this problem. The thin atmosphere and vulnerability to dust storms allows dust to remain suspended for long periods of time. Frequent dust activity can drive particles into exposed mechanical regions. Moreover, regolith acquires electrical charge through frictional interactions with rover mechanics, causing it to cling stubbornly to surfaces. Traditional solutions, such as seals and protective closings, are limited by mass constraints and the lack of sustainability across severe temperature fluctuations. As a result, dust infiltration remains as a vulnerability in rover wheel motor design.

Addressing dust infiltration in wheel motors can drastically enhance the effectiveness of the rover as a tool for exploration and scientific development, by allowing it to operate longer and reach environments that would otherwise remain unstudied.

## Information Gathering

NASA's first Mars rover, named Sojourner, "touched down" on July 4 1997, marking the beginning of Mars exploration and leading to five other similar rovers that have allowed humans to conduct research from millions of miles away. Each rover shares a common goal in the eyes of humanity: to learn, discover, and push scientific boundaries as a species. As the rovers progressed, they received upgrades, such as more instruments and better-adapted wheels, while participating in different missions. Sojourner, being the first rover, housed only a few scientific instruments, including spectrophotometers and imaging cameras, and was not very mobile. The Spirit and Opportunity rovers shared the same purpose of finding water on Mars and featured five scientific instruments as well as being much larger and faster. The Curiosity rover had a more daunting task of finding out if Mars once had what all life needs, primarily searching for the proper chemical ingredients. It housed ten different scientific instruments and weighed roughly 2,000 pounds. Finally the most recent rover that has landed, the Perseverance rover, was designed to find out if humans could ever explore mars in person and search for past or present life. It housed seven scientific instruments and weighed 2,260 lbs.

One of the biggest challenges with building a Mars rover are the conditions in which they have to operate in. All the rovers face vacuum conditions, drastic temperature fluctuations, radiation exposure, and abrasive dust particles. Of these issues, most can be simulated in a controlled environment. However, the abrasive dust particles which are present on the surface of Mars cannot be accounted for on Earth. The weather conditions on Mars make the rovers' longevity a big problem. To counter such effects, the design solution makes use of a basic scientific principle, the law of magnetism, to allow the rovers more time on Mars to collect data and send it back to Earth so that the scientists can continue to study and learn about the red planet.

## Possible Solutions

There are many ways of solving the problem that the Mars rover faces of getting dirt and other materials inside of its motors, gears, and other electronic systems that ruin the longevity of the rover and make it harder for the robot to complete its tasks on Mars. Three possible solutions that could help solve this problem to aid with scientific discovery is by using an oscillating motor, a bubble rover, and electrostatic wheels. Each of these possibilities have their advantages and disadvantages that will help the rover on its journey through Mars.

The first solution to this problem on Mars is an oscillating motor that shakes off the dirt and other materials or minerals that build on the rover using vibrations and/or a side to side motion like a four-bar mechanism. This solution would work by using gears or a rotating motor that would be placed at parts of the rover where dirt and other materials get into the rover to take them off before they become a dominant issue. This solution has many advantages such as not using too much energy, being small, not weighing too much,  and being very efficient and effective at getting rid of dirt and other small particles. In terms of complexity, this design would also be easiest to build. However, the disadvantages outweigh the advantages of this solution because it would take too many oscillating motors to get all of the necessary dirt off of the rover. It would ruin the design of the rover and cause too much drag force. Finally, this design would create too much unwanted movement from the vibrations or oscillations which could tend to move the rover off course over long periods of time. Overall, this solution on paper sounds like a good idea, however, due to the fact that it requires many motors makes the design too complicated and inefficient. Hence, it does not make this the final solution for this problem.

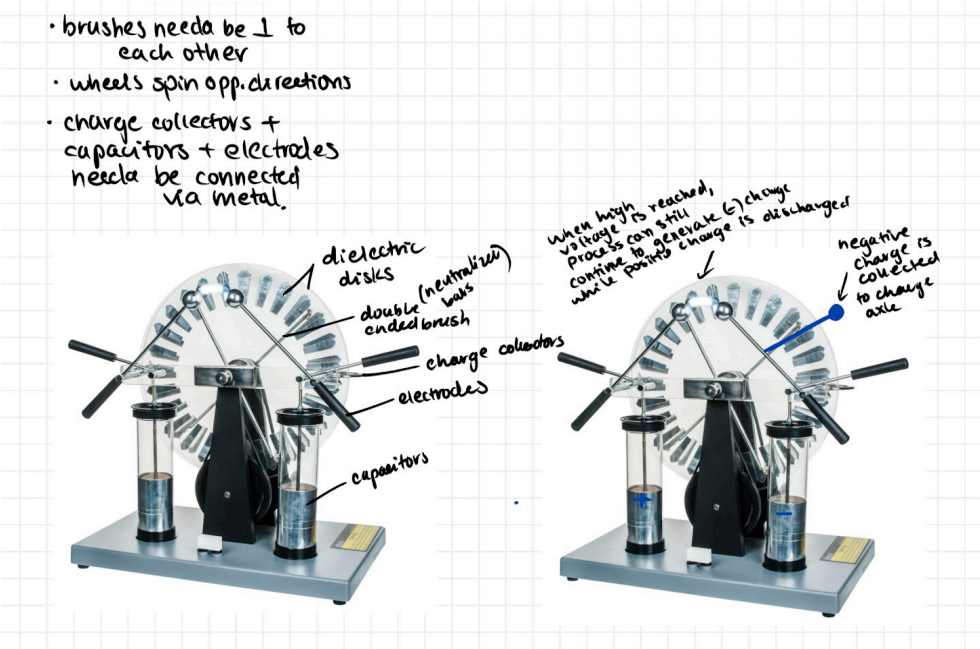
Another possible solution could be using a bubble for the rover. This solution would work as a protective bubble cover around the rover. A bubble rover could work in multiple ways such as having it enclose just the wheels or over the axles instead of the entire rover. Yet, it's important to consider that putting a bubble in any of these areas would be difficult because it would require materials that are durable, strong, and thin. The materials must be strong enough to last over long mission times, but at the same time, flexible enough to put over the rover. However, materials like this are very expensive, rare, and require lots of maintenance that cannot be attended to remotely. A bubble rover does sound like a good idea theoretically, since it can also protect the rover from bigger debris like small rocks. However, the rover would need gas to hold it up or other metals which make the rover heavier and take up more space than other solutions. The protective bubble could also interfere with the rover collecting and studying Martian samples. Even if the bubble had a few slits or holes in it, it would basically become useless because dirt and rocks could easily go through and ruin the motors or cause further damage to other parts of the rover. Overall, this solution will not be very effective because it could easily break, is too expensive, lacks materials needed to be efficient, and could interfere with rover functionality on Mars.

The third and final solution that the design team selected are the electrostatic wheels. Electrostatic wheels work by using the triboelectric principle that works when rubbing a positive and negative material together. Over time, the electrons would be displaced that generate a spark in terms of discharging electricity. This principle is best explained with the model of the Wimshurst machine. When two disks with materials from both ends of the triboelectric series are rubbed against neutralizer bars, the electrons try to move from one side to the next as the interaction occurs. While this is happening, a comb like metal piece collects these free moving electrons and stores them in capacitors until they are discharged via a spark gap. The wheels on Mars would utilize the rotational motion from the motors to generate the imbalance in charges and produce electrical sparks. These electrical sparks, depending on which end of the triboelectric series has the greater imbalance, would force any metal that comes in contact to have the same charge. The smaller dust/regolith particles on Mars are the most problematic, as they get caught in the motors, while larger particles could be useful for sampling. Since the smaller particles of Mars dust are negatively charged, this solution would utilize the triboelectric series to generate a spark that negatively charges the axle, repelling the dust particles from entering the center of the wheels.

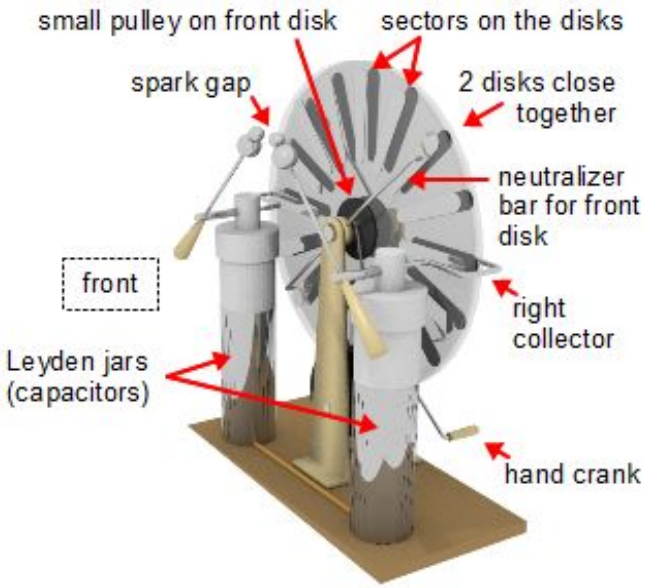
In conclusion, all of these plausible solutions would work to help solve the problem on Mars, but the best solution is the electrostatic wheels because it would be most effective and efficient in the process, provide long term protection, and it would be much easier to implement onto the rover systems. Therefore, after considering various perspectives such as the impact on the rover, effectiveness, and how the various systems would function in coordination with the rover, we realized that the electrostatic wheels would be the best solution to preventing dust accumulation on important rover systems.

## Visual Representation of Prototype/Model

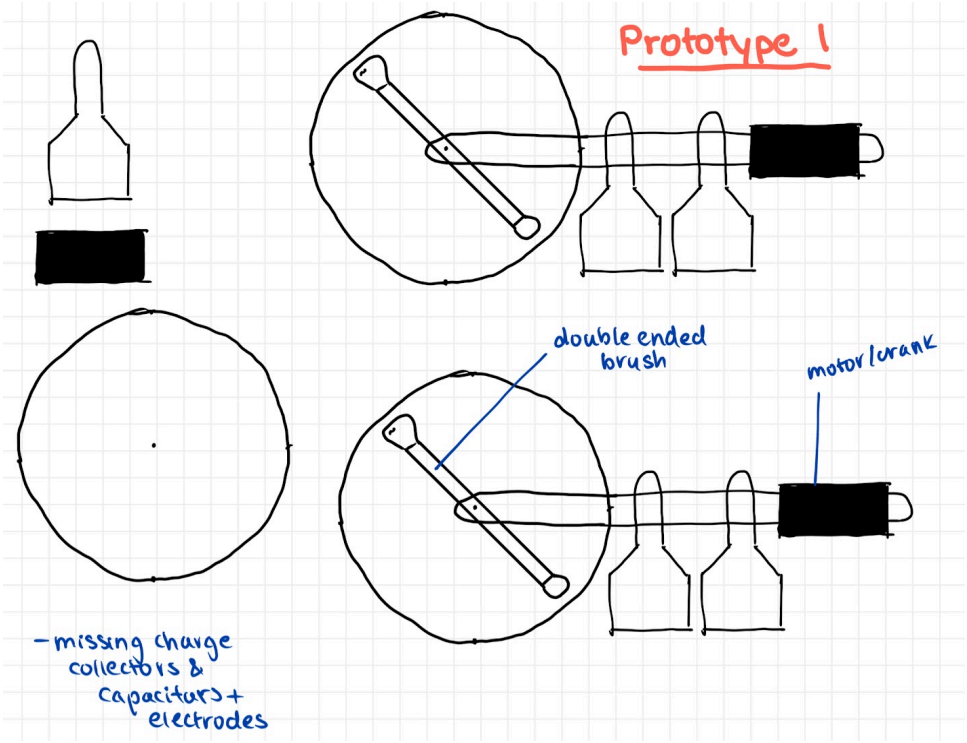
Sketches and Drawings to Learn Mechanism:



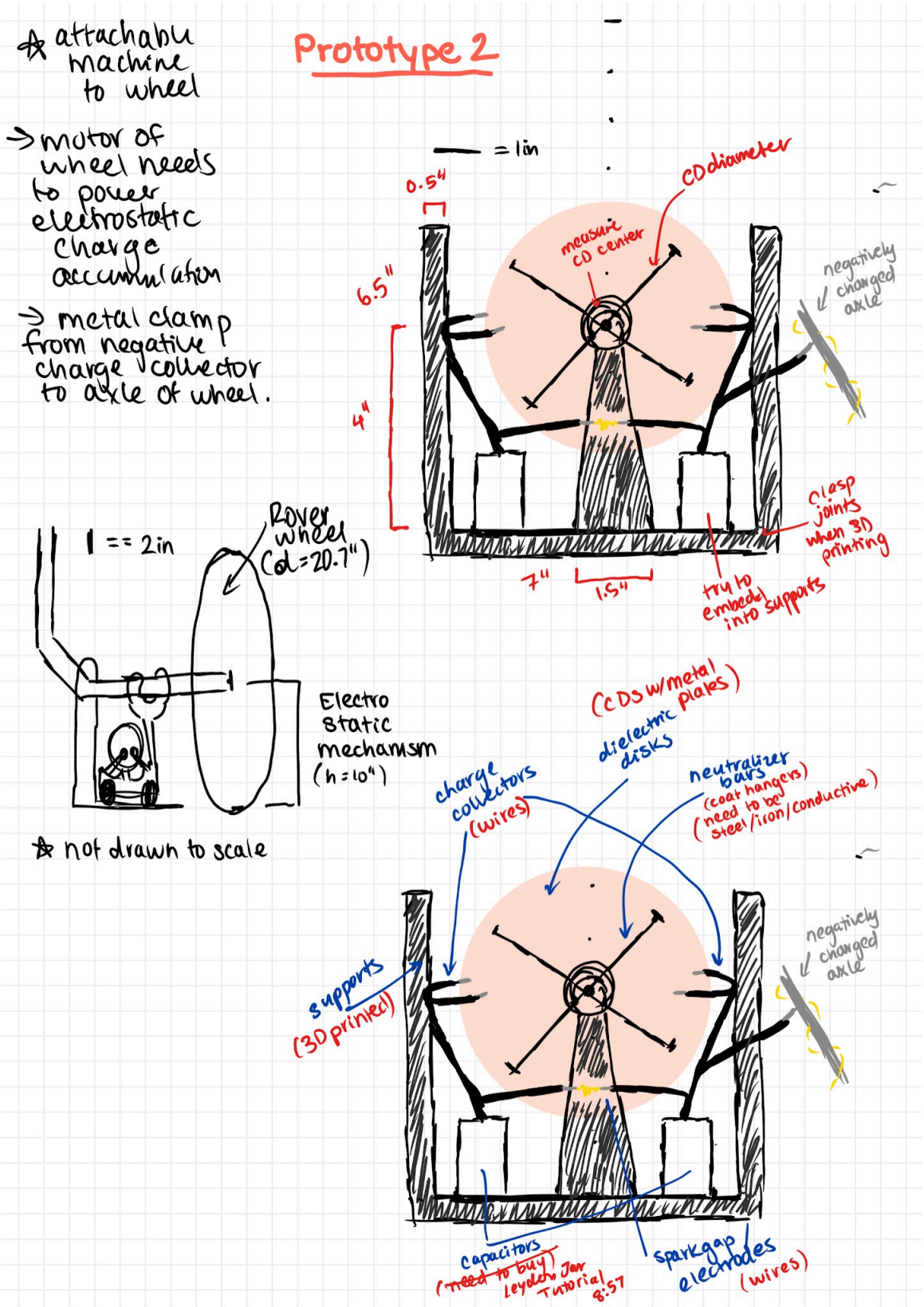

Wimshurst Machine Components Explained:



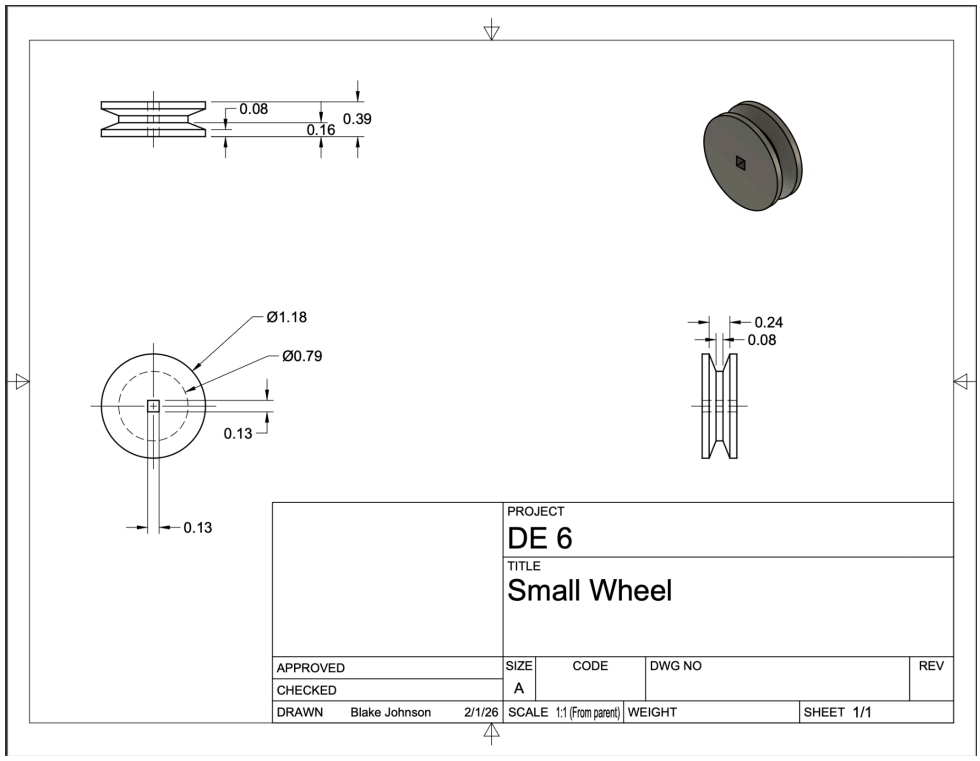
Sketches of Prototype 1 and Improvements to be Made:

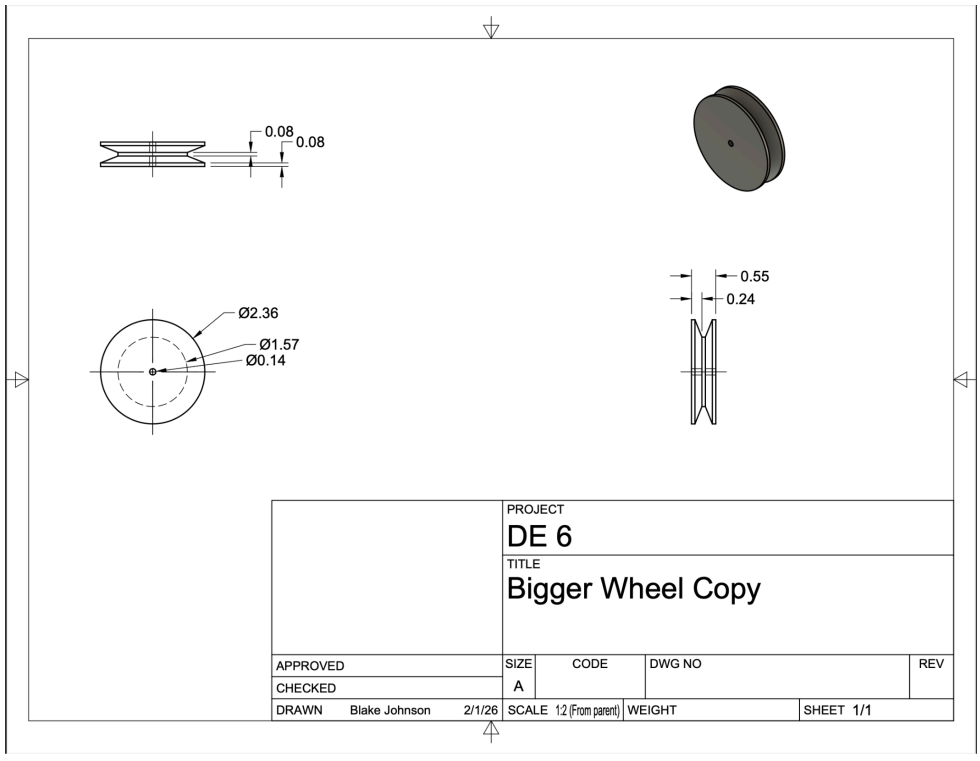


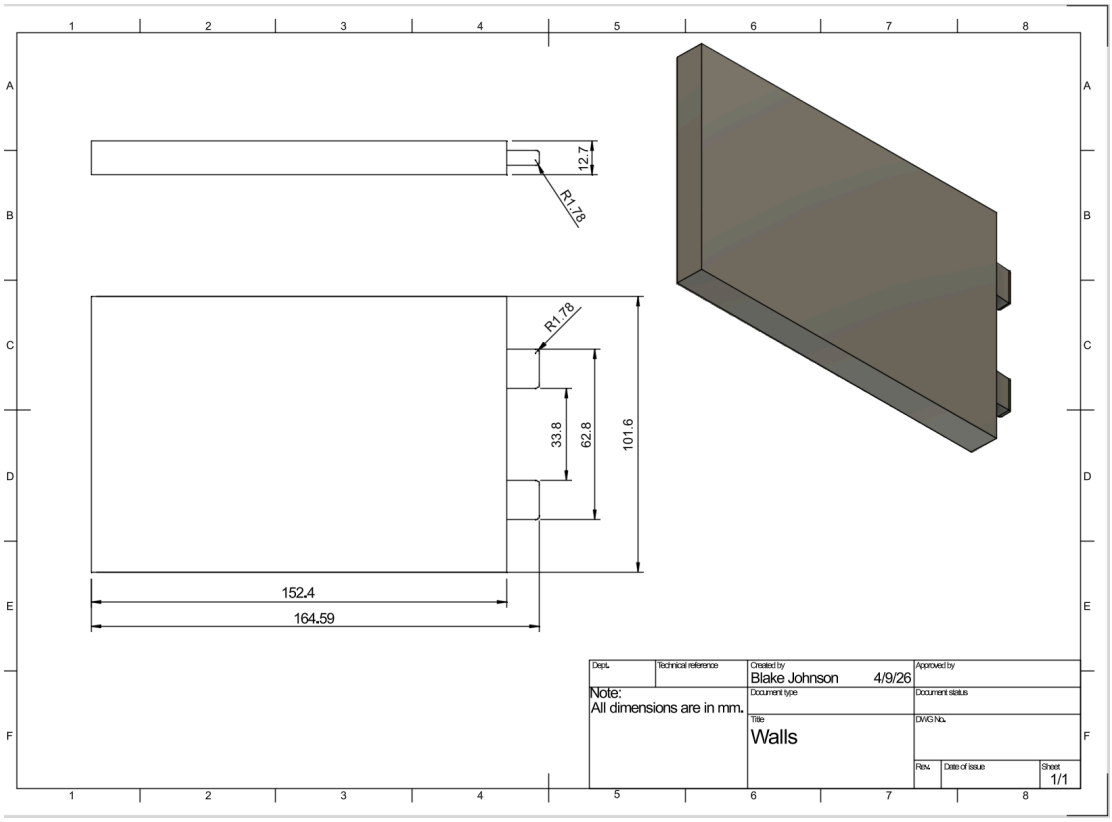
Sketches of Prototype 2 and New Plan:

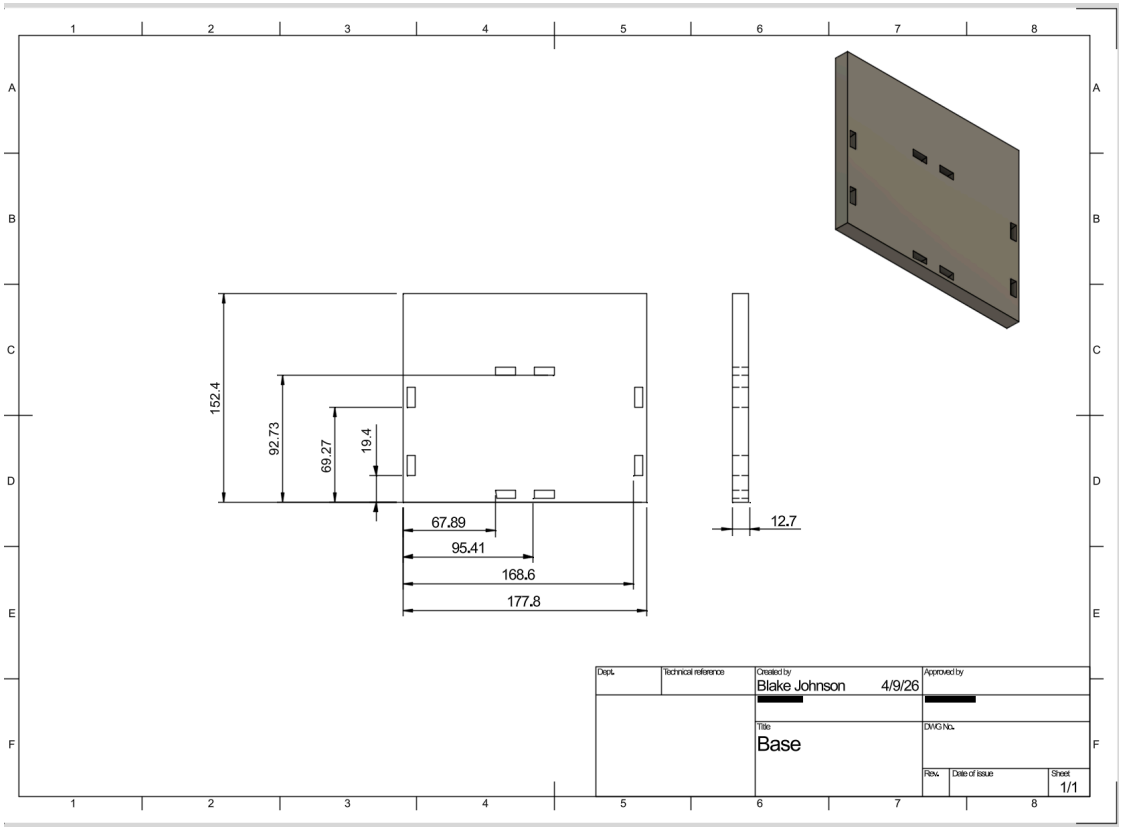


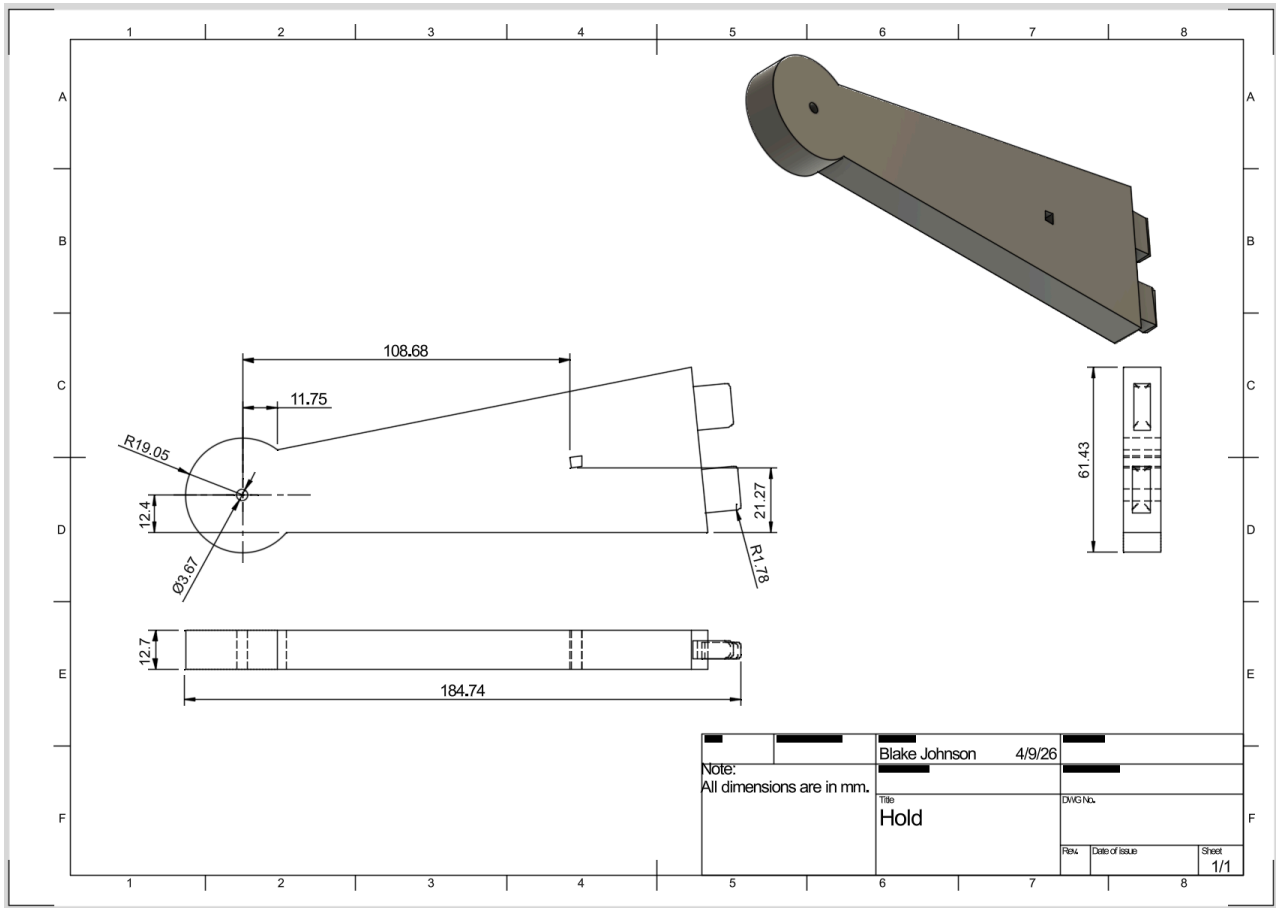
Engineering drawings for some of the components for the prototype:




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## Summary of Iteration Process

**Define the Problem:** Martian dust and sand particles pose many obstacles to the machinery and motors aboard Mars rovers. These particles can linger in the atmosphere for long periods of time thanks to dust storms called “dust devils.” Furthermore, they become electrically charged and cling more strongly to the rover than particles on Earth. How can important mechanical systems such as the axles, wheels, and motors be protected from such conditions to improve their performance and conditions in the long run?

**Target Customer:** Mars exploration agencies which build rovers for Martian terrain including NASA.

**Designer(s):** Blake Johnson, Aawash Neupane, Lakshanya Prabhu, and Jonathan Ullrich

**Criteria:**

- The design must be able to keep dust and sand particles away from the motors and the axles.
- The design must function continuously without breaks.
- The prototype must function for at least 2 minutes.

**Constraints:**

- The design must not weigh more than 5 kgs.
- The design must be scalable.
- The design should try to minimize costs.
- The design must not require another external motor but should work with the motors on the wheels of the rover.
- Solutions must make the entire process efficient thereby solving the problem.

- Technology of the solutions must be updated as needed.

**Generate Concepts:**

After many group discussions and brainstorming sessions, we came up with the following 3 ideas.

1. Oscillating Motors
2. Bubble Rover
3. Electrostatic Wheels

**Decision Matrix:**

| Solution | Energy consumption | Weight | Cost | Efficiency | Security | Total |
|---|---|---|---|---|---|---|
| **Oscillating motors** | 1 | 5 | 5 | 3 | 2 | 16 |
| **Bubble Rover** | 5 | 2 | 3 | 1 | 1 | 12 |
| **Electrostatic Wheels** | **4** | **3** | **3** | **5** | **5** | **20** |

Our conclusion was to make the electrostatic wheels since they scored the highest in the various categories of the decision matrix.

This decision matrix has a one to five scaling system. One being the worst and five the best. And one would be that it is too difficult to implement, it doesn't meet the criteria and constraints, or it doesn’t work the way it is designed to. A five would be that it can be implemented easily, meets all the criteria and constraints, and works how it is designed to. The rest of the numbers are in

between these numbers, creating a measure of design compatibility with the criteria and constraints.

The one to five scaling system is based on the criteria and constraints. The better the design follows the criteria, the better and higher number that idea will get for that category.

**Develop a Solution:**

First we came up with ideas on how we are going to approach this concept. Our main question revolved around how we would get charge on the axle without using an external source. After some research, we came upon the Wimhurst Machine which produces high voltages by converting rotational motion into electricity. We decided that this would work best for our solution since the machine can produce very high voltages in the tens of thousands of volts. The next step was determining how we could integrate this system into the wheel structure of the rover and to do this, we examined the layout of Perseverance's wheels. After doing so, we decided to encase our prototype in a shell which could protect it from the harsh conditions of Mars.

**Construct & Test and Evaluate Solution:**

<u>Prototype 1: Oscillating Motors</u>

The simplest way of preventing dust from accumulating inside the motors would be to somehow make the motors vibrate occasionally. However, while this solution would work on Earth, the electrically charged nature of Martian soil and dust means that particles attach much more firmly to the rover and simply oscillating the motor may not be enough to remove fine particles. Additionally, rapid oscillations of the motors would consume extra power and may also result in malfunctions with the motor, reducing its efficiency and lifespan. Not only could this risk the motors, but any accidents in the procedure could result in damage to the rover overall, rendering it

useless. Therefore, this prototype is a high-risk, low-reward way of approaching this problem, hence why it was scrapped after careful consideration.

Prototype 2: Bubble Rover

While this prototype could work effectively, it isn't as efficient as it would need to be large and would need to cover the entire rover. Additionally, it would require precision and many sensors which could detect Martian weather conditions and predict whether the Bubble should be deployed or not. This adds many levels of complexity and there are many points of error. Material development, especially a material which can withstand the abrasive Martian particles, would take a lot of capital and time. Therefore, while this idea could hypothetically work, the design, complexity, cost, and overall efficiency of the prototype didn't align very closely with the criteria and constraints.

Prototype 3: Electrostatic Wheels

This was the prototype that was decided on to finalize because it successfully was able to remove and keep out dirt particles that would be found on Mars, is cheaper than the other options, and was the most effective and efficient. This design works by spinning disks with aluminum, attached to wires to move the electrons around, having a positive and negative side, and attaching it to the axle to keep the negatively charged dirt particles out of the motors. Overall, this design was able to effectively keep out all of the dirt particles while still being cheap, lightweight, using easily accessible materials, easy to manufacture, and, most importantly, make the rover last longer.

**Conclusion:**

After a few prototypes and many hours of testing and designing, this final prototype worked as intended and helped solve the problem that the rovers face on Mars of getting dirt and other debris in internal components and motors which ruins the longevity of the rover and could potentially cut

its mission short. This solution and prototype meets all criteria and constraints that were needed that made it work so well and could work on Mars too. Even though this prototype isn't the best it could be and the final product because the team doesn't have the best materials and the best equipment to make, it still shows how the design would work on Mars and how it would solve the problem that the Mars rover faces. The prototype works by spinning disks with aluminum, attached to wires to move the electrons around, having a positive and negative side, and attaching it to the axle to keep the negatively charged dirt particles out of the motors. Because this prototype doesn't have the best resources and isn't the most efficient at collecting and disrupting the electrons around the design, it doesn't capture all of the needed electrons to perfectly get out all of the dirt and other particles, but it helped the team understand that the prototype, if better, could work on Mars.

## Communication of Solution

The grand engineering challenge for this year, Engineering the Tools of Scientific Discovery, aims to counter scientific barriers by designing and implementing novel tools. A historical example of this process would be the development of the electron microscope in response to the limitations of optical microscopes. Similarly, our solution with the electrostatic wheels aims to improve upon the current designs of the Mars rovers to increase their lifespan.

One of Perseverance's various discoveries was the confirmation of electrical discharges in Martian dust devils. It is hypothesized that interactions between particles in such atmospheric conditions can create various charges, with larger particles gaining positive charge while finer particles carry negative charges. The electrostatic wheels that we designed work by generating an electric charge around the axles of the rover which can be further applied to a casing surrounding the main motors and gear systems of the wheel. This is done by using a mechanism similar to a Wimshurst machine which converts mechanical energy into high voltage electricity. Through this process, the voltage generated by the machine can be applied to the wheels, creating a negative charge on the axles and the casing encompassing the motors and repelling the negatively charged fine particles on Mars. This will help to protect the various motors on rovers such as Perseverance, including the six motors on its six wheels as well as other motors throughout the rover body which help with different aspects such as moving the robotic arm.

By creating an effective method to reduce dust accumulation inside the rover machinery, our solution aims to increase the lifespan of rovers on Mars, allowing scientists to get consistent results for longer periods of time. Therefore, our design aligns with the annual theme and provides another layer of defense and our engineering process depicts the process of re-engineering the tools of scientific discovery, which in our case are the wheels of Mars rovers.

# Plan of Work Log

| TECHNOLOGY STUDENT ASSOCIATION WORK LOG | | | | |
|---|---|---|---|---|
| **Date** | **Task** | **Time involved** | **Team member responsible** (student initials) | **Comments** |
| 1. 11/4/25 - 11/5/25 | Research, brainstorm, and identify the problem | 6 hours | B.J., A.N., J. U., L. P. | Over the course of two meetings we brainstormed and created a list of possible problems that tools of scientific discovery have to improve upon them and fix those problems. |
| 2. 11/13/25 & 11/16/25 | Develop the criteria, constraints, and possible solutions | 7 hours | A.N., J. U., L. P. | Generating three possible solutions to the problem that follow the criteria and constraints: oscillating robot cleaner, bubble rover, and electrostatic wheels. |
| 3. 12/3/25 | Select our final solution | 1 hour | A.N., J. U., L. P. | Using a decision matrix (part of the design process), we selected the best possible solution to our problem: electrostatic wheels for the Mars rover. |
| 4. 12/10/25 | Finalize the details of our solution | 1 hour | B.J., A.N., J. U., L. P | We decided to make add an attachment to the Mars rover that would generate electrostatic electricity by using triboelectric principles. |
| 5. 12/16/25 | Begin coding for the motor. | 1 hour | L.P. | Used the coding environment provided by VEX to write a simple code that would allow us to control how the motor runs. |
| 6. 1/6/26 | Begin building the prototype and gathering the correct materials. | 3 days | B.J., A.N., J. U., L. P. | We asked our physics teacher about the correct materials that would allow for the triboelectric effect and got the motor. |

| TECHNOLOGY STUDENT ASSOCIATION WORK LOG | | | | |
|---|---|---|---|---|
| **Date** | **Task** | **Time involved** | **Team member responsible** (student initials) | **Comments** |
| 1/9/2026 7. | Reviewing old solution | 2 hours | B.J., A.N., J. U., L. P. | Reviewed mistakes made on old solution, tried to find room for improvement |
| 1/10/2026 8. | Brainstorming ideas, assigning roles | 4 hours | B.J., A.N., J. U., L. P. | Established the basic idea, assigned roles in the team for the smooth functioning of the project |
| 1/16/2026 9. | Modifying Main Solution | 3 hours | B.J., A.N., L. P. | Revising the primary solution to perfectly align with the problem statement. Made sure that the wheels worked. |
| 1/21/2026 10. | Planning Question format, portfolio | 4 hours | B.J., A.N., J. U., L. P. | Worked on presentation of the project as well as addressing the questions surrounding the prototype. |
| 1/26/2026 11. | Making questions and finalizing prototype | 4 hours | B.J., A.N., J. U., L. P. | Finished with the list of questions, modifying portfolio pages, |
| 1/28/2026 12. | Tri-Fold, Portfolio, and finish prototype | 6 hours | B.J., A.N., J. U., L. P. | We came up with a new Tri-fold design. Consequently updated the portfolio. |

**Resources/References**

Ashar, Unnati. “Dust Devils on Mars May Spark Lightning — Possibly Threatening NASA’s Perseverance Rover.” *Space*, 2 July 2025, www.space.com/astronomy/mars/dust-devils-on-mars-may-spark-lightning-possibly-threatening-nasas-perseverance-rover.

Atkinson, Nancy. “NASA Is Trying to Fix a Problem with One of Perseverance’s Instruments - Universe Today.” *Universetoday.com*, 2024, www.universetoday.com/articles/nasa-is-trying-to-fix-a-problem-with-one-of-perseverances-instruments.

Campbell, Stephen. “How Does a Wimshurst Machine Work?” *Demos.smu.ca*, 2025, demos.smu.ca/demos/e-n-m/43-wimshurst-machine.

Forward, Keith M., et al. “Particle-Size Dependent Bipolar Charging of Martian Regolith Simulant.” *Geophysical Research Letters*, vol. 36, no. 13, 8 July 2009, https://doi.org/10.1029/2009gl038589.

Fox, Karen. “NASA Rover Detects Electric Sparks in Mars Dust Devils, Storms - NASA.” *NASA*, 3 Dec. 2025, www.nasa.gov/missions/mars-2020-perseverance/perseverance-rover/nasa-rover-detects-electric-sparks-in-mars-dust-devils-storms/.

Marsh, Allison. “James Wimshurst’s Electrostatic Immortality.” *IEEE Spectrum*, IEEE Spectrum, 14 May 2023, spectrum.ieee.org/wimshurst-machine.

Mraz, Stephen J. “For Perseverance’s Mars Mission, Motors Are Critical.” *Machinedesign.com*, Machine Design, 19 Mar. 2021,

www.machinedesign.com/directory/mechanical-motion-systems/article/21158585/for-perseverances-mars-mission-motors-are-critical.

NASA. “Perseverance Rover Components.” *Science.nasa.gov*, NASA, 15 Nov. 2024, science.nasa.gov/mission/mars-2020-perseverance/rover-components/.

OpenStax. “18.3: Coulomb’s Law.” *Physics LibreTexts*, 24 June 2016, phys.libretexts.org/@go/page/2537.

Parkin, Chris. “Wimshurst Machine.” *Www.hsm.ox.ac.uk*, 2023, www.hsm.ox.ac.uk/wimshurst-machine.

published, Samantha Mathewson. “Tricky Mars Rocks Making Things Difficult for NASA’s Perseverance Rover.” *Space.com*, 1 July 2022, www.space.com/nasa-perseverance-mars-rover-rock-problems.

RimstarOrg. “Wimshurst Machine - How to Make Using CDs.” *YouTube*, 17 Mar. 2013, www.youtube.com/watch?v=puC6-UaT9Fk.

Smith, Yolanda. “History of the Electron Microscope.” *News-Medical.net*, 23 Aug. 2018, www.news-medical.net/life-sciences/History-of-the-Electron-Microscope.aspx.

Steltzner, Adam . “Challenges of Getting to Mars: Curiosity’s Seven Minutes of Terror - NASA Science.” *Science.nasa.gov*, 2012, science.nasa.gov/resource/challenges-of-getting-to-mars-curiositys-seven-minutes-of-terror

The Organic Chemistry Tutor. “Charging by Induction - Electrostatics.” *YouTube*, 10 Sept. 2023, www.youtube.com/watch?v=ZbBHnod_Mew.

USCB. “56.32 -- Wimshurst Machines.” *Ucsb.edu*, 2024, web.physics.ucsb.edu/~lecturedemonstrations/Composer/Pages/56.32.html.